

Sharavanan Mathivanan

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International Student with a valid Co-op Work Permit (Canada)

Education

University of Western Ontario

London, ON

BESc, Mechatronics & AI Systems Engineering

Sept. 2024 – Apr. 2029

- **Coursework:** Cyber-Physical Systems Theory, Intro to Mechatronic Design, Intro to Machine Learning, Data Engineering & Machine Learning, Algorithms & Data Structures, Circuits and Systems, Digital Systems & Signal Processing, Engineering Dynamics

Projects

Autonomous Mobile Robot (AMR v2) | ESP32, Python, PID Control

May 2026 – Present

- Custom chassis with tuned sensor placement; fuses ultrasonic and IMU data for PID straight-line control and obstacle avoidance.

Autonomous Turtle Egg Collector | Arduino, SolidWorks, Sensor Integration

Jan 2026

- Built a turtle egg conservation rover in a 3-day sprint; designed the SolidWorks mounts and wiring for ultrasonic and IR sensors.

Vision Mouse Advanced | MediaPipe, Python, OpenCV | GitHub

Jul 2025 – Aug 2025

- Hands-free eye-tracking cursor with self-calibrating blink detection that works across users, camera angles, and lighting.

RT-DETR Object Detector | PyTorch, Deep Learning | GitHub

Jul 2025 – Aug 2025

- Real-time object detection on live webcam video using a Transformer-based (RT-DETR) model, optimized for low-latency inference.

Experience

MEBOT EdTech, Robotics, AI & IT

Chennai, TN

Robotics & Embedded Systems Engineer

May 2026 – Present

- Designed and built an educational robot from scratch for K-12 learners, spanning hardware, firmware, and mechanical prototyping.
- Built a block-based IDE to program robots and sprites on ESP32, Raspberry Pi Pico, and Arduino R4 boards in multiple languages.
- Deployed both products into active use by **10+** students across MEBOT's robotics learning programs.

MAANTT Global Services

Chennai, TN

Computer Vision and AI Intern

May 2025 – Aug 2025

- Developed a Python-based monitoring system tracking **50+** concurrent inputs with **90%** detection accuracy.
- Automated reporting workflows and authored technical documentation, improving cross-team communication.

MEBOT Robotics

Chennai, TN

Robot Designer & Trainer

Mar 2024 – Aug 2024

- Designed and assembled functional robot prototypes from concept through commissioning.
- Managed project timelines and created a training curriculum for students aged 8-15.

Leadership & Design Teams

Western Aero Design

London, ON

Payload Lead

Jun 2026 – Present

- Lead electrical design and integration of the advanced-class drone payload and its mounting mechanism for reliable deployment.
- Coordinate with the software lead to integrate payload control into the drone's flight system.

Western Aero Design

London, ON

Junior Tricopter Lead

Jan 2026 – Jun 2026

- Led a cross-functional team of **5** to design, model, and manufacture UAV structural components to SAE competition requirements.
- Created SolidWorks models, drawings, and BOMs; chose materials and tolerances for 3D-printed and CNC-machined parts.

Western Aero Design

London, ON

Prototyping Engineer

Sep 2025 – Jan 2026

- Designed and 3D-printed **10+** mechanical prototypes including mounting brackets, motor housings, and fixtures.
- Engineered custom PCB layouts and wiring harnesses within one-week deadlines.

WEFIRST Robotics Club

London, ON

Software Lead, Ri3D

Jan 2026

- Led a team of **4** to build and program the robot from scratch under a **72-hour** build constraint.
- Programmed PID control loops and motor configuration for automated mechanism actuation.

Technical Skills

Languages: Python, C/C++, Java, MATLAB, JavaScript, SQL, HTML/CSS

Tools: Linux, Git/GitHub, Docker, VS Code, IntelliJ IDEA, Claude Code

Hardware: Arduino, ESP32, PCB Layout (KiCAD), SolidWorks, CAD Design, 3D Printing

AI/CV: TensorFlow, PyTorch, MediaPipe, OpenCV, Computer Vision

Certifications

CSWA, Certified SolidWorks Associate, Dassault Systèmes (2025) • **MATLAB Onramp**, MathWorks (2024)